GG-H-109B
December 10, 1975
SUPERSEDING
GG-H-109A
August 12, 1959

FEDERAL SPECIFICATION

HANGER, X-RAY FILM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hangers for processing medical and dental X-ray films in developing and drying procedures.

1.2 Classification. Hangers shall be of the following types and shall accommodate X-ray films of sizes, as specified (see 6.2):

Type I - Medical X-ray film hanger
Size 1 - 8 by 10 inches
Size 2 - 10 by 12 inches
Size 3 - 14 by 17 inches
Type II - Dental X-ray film hanger

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 66 - Steel, Chemical Composition and Hardenability.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6525

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

B 117 - Standard Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Metal. All parts of the hangers (except identification tabs for type II hangers) including clips, clip points, top bar pins, springs, frames, and any other parts, shall be made entirely of corrosion-resisting steel, conforming to the requirements for AISI 302, 303, 304, 304L, 308, or 316 of Fed. Std. No. 66.

3.1.2 Identification tabs. Identification tabs for type II dental X-ray film hangers, shall consist of a suitable material with a white matte finish which will readily accept pen and pencil marks and which will not show any deterioration when subject to the chemical resistance test specified in 4.4.2.

3.2 Design and construction, type I. Each type I hanger shall consist of a top bar as specified in 3.2.2, a rectangular frame, spring clips, and, if required, a crossbar designed and constructed to hold film in a taut, vertical position for processing.

3.2.1 Sizes. Hangers shall be furnished in sizes to accommodate X-ray films of the sheet sizes specified in 1.2.

3.2.2 Top bar. The top bar shall be 16 ± 1/16 inches long of 1/4-inch channel or solid bar stock. If the top bar is a channel, it shall be formed of steel 0.031 inch thick with commercial tolerances. The top bar shall be 0.250 ± 0.015 inch by 0.250 ± 0.015 inch. Each leg of the channel shall form an angle of 90°, +5°, -0° with the base. The sides of the top bar shall be parallel with the plane of the wire frame, with a maximum offset of 0.005 inch.

3.2.3 Top bar pin. For sizes 1 and 2, there shall be a pin 5/8 inch ± 1/8 inch long attached to the under surface of one end of the top bar 3/4 inch ± 1/8 inch from one end and 14-3/8 inches ± 1/8 inch from the opposite side of the frame, outside-to-outside dimensions. The other end of the top bar shall extend 3/4 inch ± 1/8 inch beyond the frame.

3.2.4 Frame. For hanger sizes 1, 2 and 3, the frame shall be constructed from 3/16 inch diameter rod (commercial tolerances permitted), securely riveted or welded to the top bar, and shall be formed to hold film with its long axis vertical. To assure complete tank submersion of film secured in the hanger, the design for sizes 1 and 2 shall be such that the upper clips fall at least 3 inches below the top bar with a film mounted into position. For size 3 hanger, the two top spring clip assemblies shall be securely spot welded to the under-surface of the top bar. Dimensional requirements shall be as indicated in table I.

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TABLE I. Frame, dimensional requirements

Size	Overall vertical height (inches)	Inside width of frame (inches)	Outside width of frame (inches)
1	14-7/4 minimum	8 minimum	-----
2	16-7/16 minimum	10 minimum	-----
3	19-7/16 ± 1/16	14 minimum	14-1/2 maximum

3.2.5 Clips. Four spring clips shall be riveted, clinched, or welded to the frame to securely grip film with square or round (3/8-inch radius) corners, not less than 3/16 inch from the nearest film edge. The two bottom clips shall be attached to the frame. The two top clips shall be attached to springs of sufficient tension to hold the film uniformly taut without warping the film or without pulling the film out of the clips in processing. The design of the clips shall be such that trapping of solution is minimized. The overall thickness of the clip shall be no greater than 1/2 inch. Clips shall exhibit no sign of distortion or loss of spring action after being tested as specified in 4.4.5.

3.2.6 Joints. All joints shall be securely riveted, spot-welded, or clinched. All welds shall be ground smooth with an iron-free grinding wheel.

3.2.7 Identification marking. The size of the hanger and the name or registered trademark of the manufacturer shall be stamped on the top bar of all hangers.

3.3 Design and construction, type II. Type II, dental X-ray film hanger, shall be designed to securely hold 14 dental X-ray films, 7 on either side of a vertically placed supporting hanger. It shall consist of a supporting hanger, as specified in 3.3.2 and 7 pairs of film clips, as specified in 3.3.3. Film clip pairs shall be equally spaced on the supporting hanger with a minimum of 1-9/16 inches between horizontal centerlines. The horizontal centerline of the top pair of film clips shall be at least 3-1/4 inches below the top of the hanger and the horizontal centerlines of the lowest pair of film clips shall be at least 7/8 inch from the bottom of the hanger. Unless otherwise specified, a generous tolerance of 1/32-inch (non-cumulative) shall apply to each dimension of the type II hanger. Figure 1 illustrates one acceptable, but not the only acceptable configuration.

3.3.1 Dimensions. The overall height of the hanger shall not exceed 14 inches; the overall width of the hanger shall not exceed 2-3/8 inches.

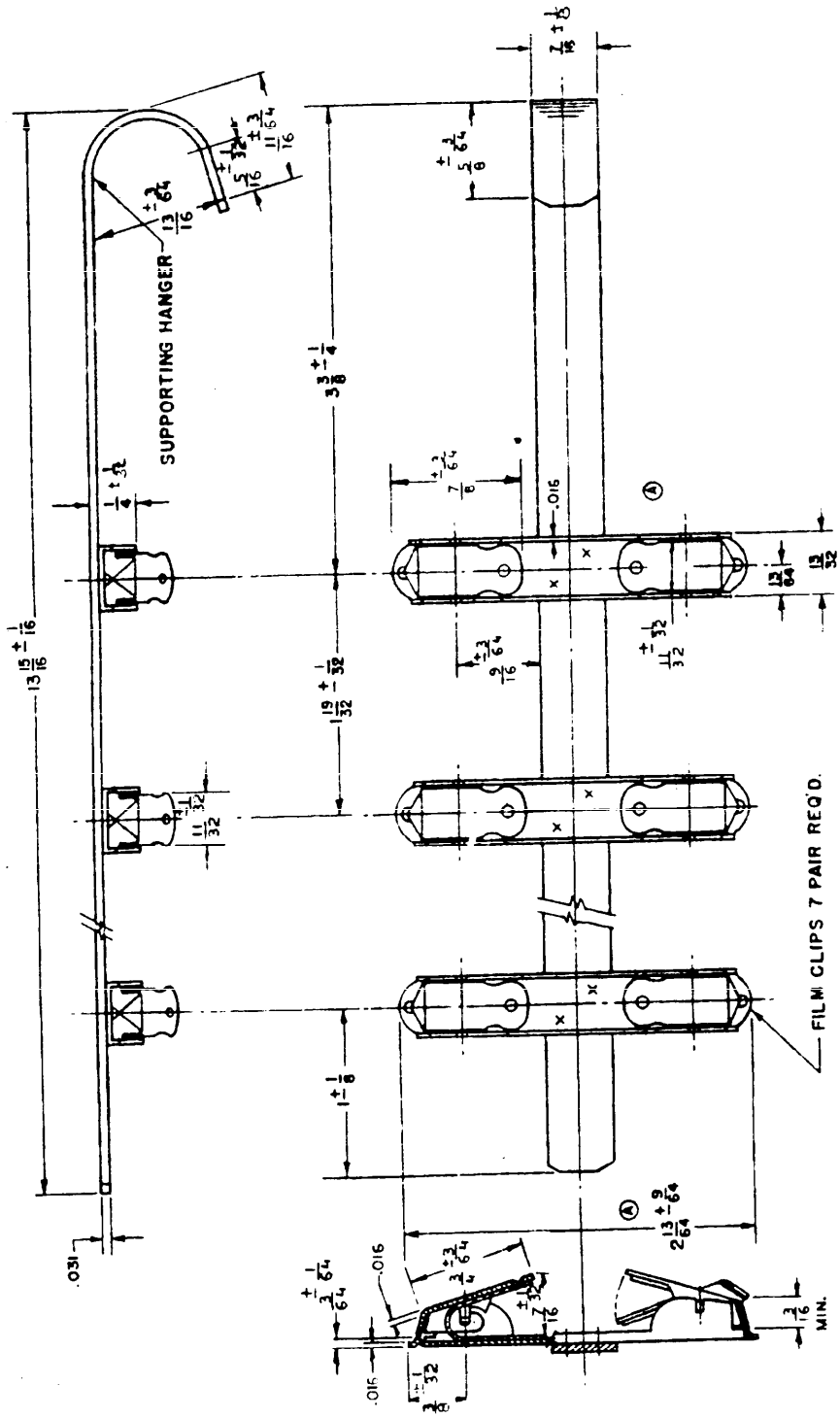
3.3.2 Supporting hanger. Supporting hanger shall consist of a straight bar or strip of steel 0.031 inch thick, with commercial tolerances, curved at one end to permit suspension over the edge of a developing tank.

3.3.3 Film clips. **The film clips shall be firmly and securely fastened in pairs, transversely, to the supporting hanger by welding to provide 7 clips on each side of the hanger. The pairs shall be spaced as shown in figure 1. The clips shall be flat spring type of suitable size and design, and shall be so placed on the hanger that films will be held away from the sides of the tank at a sufficient distance to allow free circulation of the surrounding liquid. Each clip shall securely grip the film in a manner to minimize crimping and shall be capable of easily releasing the film when desired. Clips shall exhibit no sign of distortion or loss of spring action or failure to grip film securely after being tested as specified in 4.4.5.**

3.3.4 Film-clip bite. Each film clip shall have a maximum bite of 0.118 inch.

3.3.5 Identification tabs. Dental X-ray film hanger shall have three removable white matte finish tabs that will slide onto the supporting hanger.

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ALL DIMENSION IN INCHES

FIG.-1
TYPICAL CONFIGURATION OF HANGER, X-RAY FILM PROCESSING, DENTAL

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3.3.6 Marking for identification. Dental X-ray film hangers shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.4 Spot-welding. All spot welds shall be sufficiently strong that when tested in shear or bending, as specified in 4.4.4, the material adjacent to the weld will fail before any rupture of the weld is apparent.

3.5 Passivation. The hanger shall be treated to restore corrosion-resisting properties after fabrication. Type I and type II hangers shall pass the chemical resistance test specified in 4.4.2. Type II hangers shall pass the salt spray test specified in 4.4.3. A corrosion spot visible to the naked eye, any pitting, or etching of any surface shall be cause for rejection.

3.5.1 Electropolishing. As an alternate to passivation, (3.5) the hangers may be electropolished. Electropolished hangers shall be subject to the same criteria as passivated hangers after being tested for chemical resistance (4.4.2). Electropolished type II hangers shall also be tested with the salt spray test (4.4.3).

3.6 Workmanship. Workmanship shall be first class throughout. Hangers shall be free from defects which detract from their appearance or may impair their serviceability. Finish shall be smooth and free of pits or corroded areas. Dimples, caused by spot welding, shall not be cause for rejection of type II hangers, provided there is no burning through or discoloration of the metal and no sharp or raised area. All joints shall be securely and properly fastened. Edges and corners shall be free of burs or sharpness. Component parts of the finished hanger shall be free of cracks and unspecified bends.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table II.

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TABLE II. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	S-4	1.0
Total defects (Major and Minor combined)	S-4	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, for an inspection level of S-2, and table III. Acceptance number shall be zero for tests.

TABLE III. Sampling for tests

	Characteristic	Requirement	Test procedure
Hanger	Chemical Analysis	3.1	4.4.1
	Chemical Resistance	3.1.2	4.4.2
	Welding	3.4	4.4.4
Hanger (type II)	Salt spray	3.5, 3.5.1	4.4.3
Clips	Fatigue	3.2.5, 3.3.3	4.4.5

4.2.3 Raw materials. Four ounces of each lot of each metal used in the construction of the hangers shall be selected for chemical analysis.

4.3 Examination. The hangers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in table IV. Examination shall not be restricted to the classified possible defects listed below.

TABLE IV. Classification of defects

Major	
101	Clips do not uniformly and securely grip film placed in hanger.
102	Film clip bite for type II hangers exceeds 0.118 inch.
103	Hanger not complete with all specified component parts.
104	Riveted joints not securely and properly fastened.
105	Clinched joints not securely and properly fastened.
106	Spot-welded joints not secure and free of scale, flux, cracks, or discoloration.
107	Component parts in finished hanger not free of cracks and unspecified bends.
108	Edges and corners of components in finished hanger not free of burs or sharpness.
109	Finish not smooth and free of pits or corroded areas.
Minor	
201	Identification marking not present or not permanent and legible.
202	Unit not clean, not free of dirt or extraneous matter.

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4.4 Tests.

4.4.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with the chemical analysis or the Spectrochemical Analysis test requirements of Fed. Test Method Std. No. 151.

4.4.2 Chemical resistance test.

4.4.2.1 Solutions. Prepare developing and fixing solutions in accordance with the formulas given in table V and table VI.

TABLE V. Developing solution

Chemical 1/	Amount
Distilled water, 50°C ± 5°C	500 ml
Monomethyl-para-aminophenol sulfate	2.2 grams
Sodium sulfite (desiccated)	72.0 grams
Hydroquinone (1,4-dihydroxybenzene)	8.8 grams
Sodium carbonate (monohydrate)	56.0 grams
Potassium bromide	4.0 grams
Distilled water to make	1.0 litre

1/ Dissolve chemicals in the order given. Each chemical should be completely dissolved before the next one is added. Use without dilution.

TABLE VI. Fixing solution

Chemical 1/	Amount
Distilled water, 50°C ± 5°C	600 ml
Sodium thiosulfate, anhydrous	225.0 grams
Sodium bisulfite, anhydrous	10.0 grams
Boric acid (crystals)	10.0 grams
Sodium acetate, anhydrous	30.0 grams
Aluminum potassium sulfate (alum)	30.0 grams
Distilled water to make	1.0 litre

1/ Dissolve chemicals in the order given. Each chemical should be completely dissolved before the next one is added. Use without dilution.

Clean two sample hangers thoroughly by a method that will assure removal of all grease and oil. During the tests, the temperature of the solutions shall be maintained at 43° ± 3°C.

4.4.2.2 Developing solution test. Immerse one hanger to one-half its height in the developing solution for 72 hours, remove, and permit the excess solution to drain off (do not rinse or dry). Place a sheet of unfixed X-ray film of appropriate size in the hanger and suspend hanger and film in the developer to such depth that the liquid level is 1/4 inch above the top of the film. Cover the container so as to confine vapors and maintain at a temperature 43° ± 3°C for 72 hours. Discard the film, rinse, dry, and examine the submerged portion of the hanger for pitting or etching and evidence of corrosion.

4.4.2.3 Fixing solution test. Take the second hanger and duplicate the test described in 4.4.2.2, using the fixing solution and a sheet of film that has been exposed to white light.

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4.4.3 Salt-spray test. One hanger shall be randomly selected from the lot offered for acceptance. The hanger shall be tested for salt spray for a period of 24 hours in accordance with ASTM B 117.

4.4.4 Spot-weld strength. One hanger shall be randomly selected from the lot of material offered for acceptance. Parts joined by spot-welding shall be mechanically stressed until failure occurs. Welds shall then be examined for compliance with 3.4.

4.4.5 Clip fatigue test. Two hangers shall be randomly selected from the lot offered for acceptance. Three clips on the type I or type II hanger shall also be randomly selected. Each of the three selected clips shall be opened fully and allowed to close 100 times. Clips shall be examined for distortion or loss of spring action after test is completed.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table VII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE VII. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. One hanger of the type and size specified shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. Six unit packages containing hangers of one type and size only shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four intermediate containers containing hangers of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Four intermediate containers containing hangers of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure of the box shall conform to method I or II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hangers are intended for use in processing medical and dental X-ray film in developing and drying procedures.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of hanger and size of X-ray film to be accommodated (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following items appearing in the Federal Supply Catalog:

Stock No.	Item Identification	Specification Designation Type	Size
6525-00-603-1760	HANGER, X-RAY FILM, Processing, 8 by 10 inches	I	1
6525-00-603-1765	HANGER, X-RAY FILM, Processing, 10 by 12 inches	I	2
6525-00-603-1775	HANGER, X-RAY FILM, Processing, 14 by 17 inches	I	3
6525-00-603-1795	HANGER, X-RAY FILM, Processing, Dental	II	--

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6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

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VA-DMS
GSA-FSS
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